Naimisha Suguru

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PROFESSIONAL SUMMARY

Versatile and results-driven **AI Engineer** with 3+ years of experience in full-stack and backend development, now specializing in **LLMs, RAG pipelines, vector search, and semantic retrieval systems**. Possesses a strong engineering foundation built through hands-on work with Python (FastAPI), Node.js, and Java (Spring Boot), combined with practical experience integrating advanced AI capabilities using **LangChain, embeddings, vector databases, and OpenAI/Azure OpenAI APIs**. Adept at designing and implementing modular, reliable, and cloud-ready application components that support intelligent automation and AI-driven workflows.

Demonstrated ability to operate across the complete **software development lifecycle**, from API design, data processing, and feature implementation to deployment, testing, evaluation, and continuous optimization of AI-driven features. Known for adopting **clean architecture principles**, reusable component patterns, and maintainable engineering practices that improve overall system quality and scalability. Experienced in applying **prompt engineering**, model evaluation techniques, and retrieval strategies to improve the accuracy, latency, and reliability of AI-powered applications.

Well-versed in **CI/CD workflows**, containerization, cloud deployments, and modern DevOps practices, ensuring seamless integration of AI features into production environments. Collaborates effectively with cross-functional teams, translating ambiguous requirements into structured technical solutions and ensuring smooth delivery in agile settings.

Aiming to progress into a Senior AI/ML and LLM Engineering role within the next five years, strengthening expertise in retrieval architectures, scalable model integration, distributed AI systems, and cloud-native development. Committed to building robust, production-grade AI systems that combine technical depth with practical, real-world impact.

TECHNICAL SKILLS

Programming Languages/ Databases	Python, Django, Django REST Framework, Flask, React.js, Next.js, Redux, JavaScript, TypeScript, HTML5, CSS3, Bootstrap, MySQL, SQL Server, PostgreSQL, MongoDB
AI & LLM Technologies	LangChain, OpenAI/Azure OpenAI APIs, Embeddings, RAG Pipelines, Vector DBs (FAISS, MongoDB Atlas), Semantic Search
Big Data Technologies	AWS (EC2, S3, IAM, RDS, CloudWatch)
Other Tools	JIRA, GitHub/GitLab, Alteryx, Jupyter Notebooks, PyCharm,
System Design & Architecture	RESTful APIs, Microservices, JWT Authentication
Development Methodologies	Agile, Scrum
Messaging & Integration	RabbitMQ, Apache Kafka, AWS SQS, Celery, Redis (Pub/Sub), WebSockets

WORK EXPERIENCE

MICROGIG INC | Dallas, TX **Nov 2025 - Present**
Role: Software Engineer

Key Responsibilities:

- Implemented different aspects of **React JS** by creating interactive UI's supporting One-way data flow, Virtual DOM, JSX concepts.
- Developed and integrated **LLM-powered backend services** using Python and FastAPI for internal AI-driven workflows.
- Implemented components of **Retrieval-Augmented Generation (RAG)** including embeddings, document preprocessing, and retrieval logic using LangChain.
- Built and optimized **semantic search and vector search pipelines** using vector databases such as FAISS and MongoDB Atlas.
- Designed and evaluated **prompt engineering strategies**, conducting latency tests, accuracy checks, and output consistency analysis.
- Containerized AI microservices with Docker and supported **cloud-native deployments** on AWS.
- Worked closely with product and engineering teams to identify **AI feature opportunities**, performance improvements, and user-impact metrics.
- Enhanced system reliability through **structured logging, monitoring, and observability practices** for LLM-driven components

Environment: Python, FastAPI, LangChain, OpenAI/Azure OpenAI, Embeddings, FAISS, MongoDB Atlas Vector Search, Node.js, RESTful APIs, React.js, Docker, AWS (EC2, Lambda, S3), GitHub Actions, Linux, Postman, PyTest

Accenture | Hyderabad, India **Aug 2021 – Jul 2023**
Client: Kaiser Permanente
Role: Custom Software Engineer

Key Responsibilities:

- Developed **React JS** applications by completing stories in a timely manner while maintaining best practices outlined by the team lead.
- Worked on React JS Virtual Dom, react views and rendering UI using different custom HTML Tags components.
- Implemented different styling components across application modules using **Bootstrap**, **CSS, Styled Components, SASS, Inline JSX styles, Styles-JSX, Web, Material-UI** for enhancing user interface, responsiveness, and styling.
- Used Bootstrap & Materia**l** for enhancing different selection controls like drop down menus, toggles, and search bars on client-side application interface.
- Responsible for UI and architecture by building components using libraries including Tree, Slide-View, and Table Grid.
- Worked with the **UX/UI** designers, to make the screens more user friendly, as per the need and requirements of the customers.
- Built major backend modules (auth, payments, media handling) using Django and integrated **PostgreSQL**.
- Managed ORM models, custom serializers, and API versioning in **Django REST** Framework.
- Used **NGINX** and **Gunicorn** for serving Django apps in Dockerized environments.
- Configured **AWS Elastic Beanstalk (EBS)** for scalable deployment.
- Created management commands and automated scripts for data cleanup and scheduled tasks.
- Ensured code coverage with **Selenium** for end-to-end flows and **PyTest** for backend logic.
- Designed and implemented a machine learning algorithm to optimize logistics operations using **Python** and **Pandas** for data preprocessing and model integration.
- Extracted large datasets from a **SQL** database, converted them into pandas Data Frames, and performed comprehensive data cleaning and transformation for accurate model input.
- Optimized data retrieval and computational workflows to efficiently handle and process large-scale datasets, reducing processing time significantly.
- Automated the generation of results by exporting one set as **Excel** reports using **openpyxl/xlsxwriter** and storing another set back into the database.
- Integrated **RabbitMQ** to manage and queue multiple ad-hoc logistics requests, ensuring scalability and asynchronous processing.
- Initially implemented the system for one state and successfully scaled it to PAN-India operations, maintaining consistent performance across regions.
- Set up email notifications using **smtplib** and **Celery** to alert stakeholders about processing status and key operational updates.

Environment: React JS, Bootstrap, CSS, Styled Components, SASS, Inline JSX styles, Styles-JSX, Web, Material-UI, UX/UI, PostgreSQL, NGINX, Gunicorn, AWS EBS, Selenium, PyTest, Python, Pandas, SQL, Excel, openpyxl/xlsxwriter, RabbitMQ, Celery, Smtplib

XchangeTechs Star Alliance LLP | Bangalore, India **Sep 2020 – Jul 2021**
Role: Associate Software Developer Intern

Key Responsibilities:

- Designed and implemented the frontend of the application to ensure an intuitive and user-friendly interface using **Next JS**, **React JS / Node JS / MongoDB**
- Integrated **Redux** to efficiently manage the application state and facilitate seamless data flow.
- Leveraged Redux store, actions, and reducers to handle complex data management and application logic.
- Implemented Redux middleware, such as **Redux Thunk** or **Redux Saga**, to handle asynchronous operations, such as API requests. by providing a useful tool for debugging and monitoring the application's state changes.

- Utilized NPM packages such as **Axios** to handle HTTP requests and interact with APIs for fetching real-time data.
- Integrated NPM packages like **moment.js** to efficiently handle and format date and time-related functionalities in the application.
- Leveraged NPM packages such as **react-router-dom** to implement routing and navigation within the application, ensuring smooth transitions between different pages and components.
- Suggested required efficient Software Design for projects.
- Migrated legacy **Django** templates to **React SPA** for admin and reporting modules.
- Built dynamic and reusable React components with form validations using **Formik** and **Yup**.
- Integrated React Router for dynamic route rendering and component switching.
- Used Axios for communicating with **Django** APIs and handled global API errors via middleware.
- Worked on integrating **Redis** with **Python** applications for caching and real-time data handling.
- Implemented request logging in Django to securely store incoming **JSON** data in the database for auditing and security tracking.
- Implemented salted hashing in a Django application to securely store and verify sensitive user information such as passwords and API tokens, enhancing protection against brute-force attacks.
- Created responsive layouts using **Bootstrap** and custom **CSS** modules.
- Maintained state via **React Context API** before transitioning to Redux.
- Developed and maintained models, views, and templates using Django ORM.
- Used **Celery** with **RabbitMQ** to offload time-intensive processes like report generation and mail jobs.
- Managed user roles and access permissions using Django's built-in auth system.
- Wrote shell scripts and **Docker** Compose files for setting up dev/staging environments.
- Applied server-side rendering (**SSR**) for faster UI rendering in high-load admin apps.
- Wrote utilities and helper modules for backend logic reuse.

Environment: Next JS, React JS, Node JS, MongoDB, Redux, Redux Thunk, Redux Saga, Axios, moment.js, react-router-dom, React SPA, Formik, Yup, Redis, Python, Django, JSON, Celery, RabbitMQ, Docker, SSR

PROJECTS

Ad Copy Generator | *Angular, Spring Boot, Firebase, AWS, OpenAI API* **| GitHub**

- Built a full-stack **AI-powered ad copy generator** for small businesses.
- Integrated **OpenAI GPT-4 & DALL·E** for generating social media ad content and images.
- Developed **Angular frontend** with **Google Sign-In authentication via Firebase**.
- Hosted frontend on **AWS S3 + CloudFront**, backend on **EC2**, and data in **MongoDB**.
- Delivered a production-ready, cloud-native application with monetization potential.
- Implemented usage analytics and conversion tracking to optimize ad-generation workflows and drive user retention.

Smart Meal Generator | *React.js, Node.js/Express, Firebase Auth, AWS, GPT-4 API* | GitHub

- Designed an **AI-based meal recommendation app** that generates personalized meal plans.
- Integrated **Nutrition APIs (USDA/Nutritionix)** to calculate ingredient calories.
- Supported **dietary preferences** (vegetarian, vegan, gluten-free) with precise portioning.
- Developed responsive **React frontend**, backend in **Node.js/Express**, and hosted via **AWS Amplify**.
- Implemented **Google OAuth with Firebase Authentication** for secure access.

Snow-Mow Solutions | *React.js, Node.js, MongoDB* | GitHub

- Built a **web platform for snow removal & lawn mowing services**, connecting customers with providers.
- Implemented **multi-role system** (Admin, Service Provider, Customer) with role-based workflows.
- Added features for **location-based provider search, quotations, and payments with commission model**.
- Developed service history, reviews, and rating system to improve transparency

EDUCATION

University of Central Missouri *Warrensburg, MO*
Master of Science, Computer Science **Aug 2025**
Jawaharlal Nehru Technological University (JNTU) *Hyderabad, India*
Bachelor of Technology **Sep 2021**